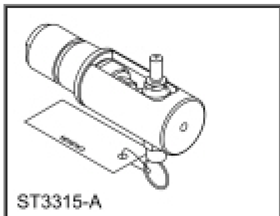
Leakage Inspection

Ford Ecat Electronic Parts Catalog

Special Tool(s)

	Leak Tester, Torque Converter 307-691

Material

Item	Specification
Dye-Lite® ATF/Power Steering Fluid Leak Detection Dye 164-R3701 (Rotunda)	—

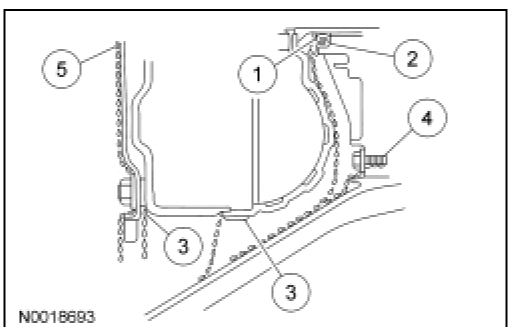
Leak Check Test

NOTE: *When diagnosing transmission leaks, the source of the leak must be positively identified prior to repair. If the vehicle is driven extensively between adding the fluorescent additive and performing the leak test, the leaking oil can spread and make identifying the location of the leak difficult.*

1. Clean off any transmission fluid from the suspected leak area.

2. Add Dye-Lite® ATF Power Steering Fluid Leak Detection Dye to the transmission fluid. Use one 30 ml (1 fl. oz) of dye solution for every 3.8 L (4 qt) of transmission fluid.

3. Start the engine and drive at least 1 mile to engage the TCC . Using a black light, observe the suspected leak area.



4. If the source of the leak is obvious, repair as required. Leaks from the torque converter housing can originate from several locations. The paths which the fluid takes to reach the bottom of the torque converter housing are shown in the illustration. The 5 steps following correspond with the numbers in the illustration.
   1. Transmission fluid leaking by the converter hub seal lip will tend to move along the drive hub and onto the back of the torque converter. Except in the case of a total seal failure, transmission fluid leakage by the lip of the seal will be deposited on the inside of the torque converter housing only, near the outside diameter of the housing.
   2. Transmission fluid leakage by the outside diameter of the converter impeller hub seal and the case will follow the same path that leaks by the ID of the converter hub seal follow.
   3. Transmission fluid leakage from the converter cover weld or the converter-to-flexplate stud weld will appear at outside diameter of torque converter on the back face of the flexplate and in the converter housing only near the

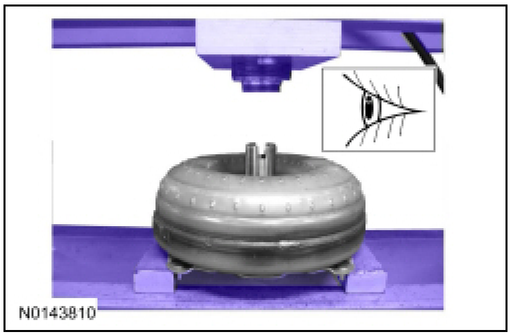
    flexplate. If a converter-to-flexplate lug, lug weld or converter cover weld leak is suspected, remove the converter and pressure check.
    4. Transmission fluid leakage from the bolts inside the converter housing will flow down the back of the torque converter housing. Leakage may be from loose or missing bolts.
    5. Engine oil leaks from the rear main oil.

5. Remove the torque converter.

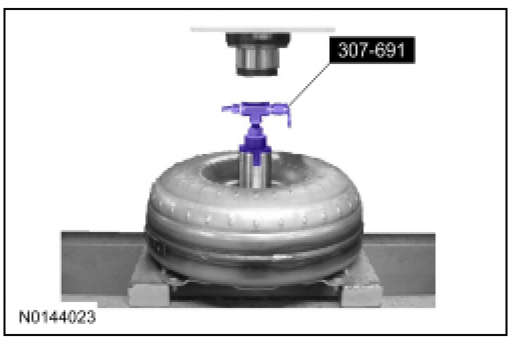
6. Using a black light, observe the torque converter housing. Inspect for evidence of dye from the pump bolts, pump seal, and torque converter hub seal. Repair as required.

7. If the source of the leak is not evident, continue with this procedure to leak test the torque converter.

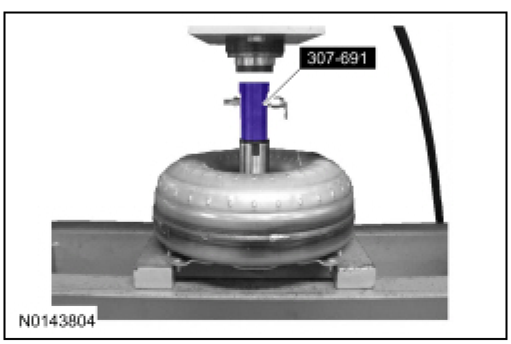
8. Install the torque converter in the arbor press. Support the torque converter on the mounting pads.



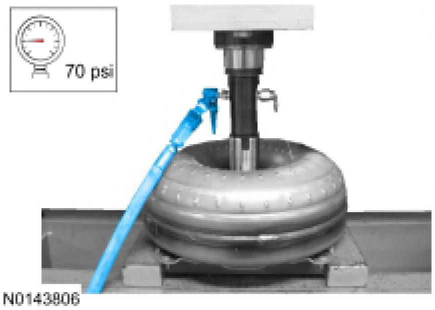
9. Install the Leak Tester, Torque Converter 307-691 into the torque converter hub. Tighten to 25 Nm.



10. Secure the press. Only apply enough force from the press to seal the Leak Tester, Torque Converter 307-691 into the torque converter hub.



11. Connect a compressed air supply to the Leak Tester, Torque Converter 307-691.



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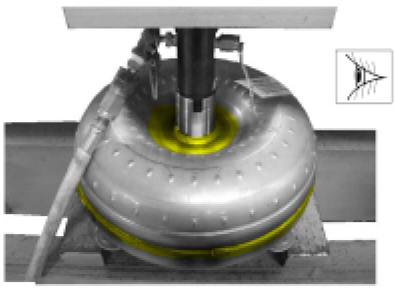
12. With air pressure applied to valve, inspect for leaks at the converter hub weld and seams. A soap bubble solution can be applied around those areas to aid in the diagnosis. If any leaks are present, install a new torque converter.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13. With air pressure applied to valve, inspect for leaks at the stud or mounting pad and balance weight welds. A soap bubble solution can be applied around those areas to aid in the diagnosis. If any leaks are present, install a new torque converter.


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14. After leaks are repaired, clean the remaining transmission fluid dye from serviced areas.